

Machine Learning-Based Prediction of Compressive Strength in Cu-Fe High Entropy Alloys: A Comparative Study

Abstract

High entropy alloys (HEAs) represent a revolutionary class of materials with exceptional mechanical properties, but predicting their compressive strength remains challenging due to complex compositional and microstructural interactions. This study presents a comprehensive machine learning approach to predict compressive strength in Cu-Fe based HEAs using a merged dataset of 305 samples with 81 engineered features. We evaluated four regression models: Random Forest, XGBoost, Support Vector Regression (SVR), and an ensemble approach. XGBoost achieved the best performance with a test RMSE of 335.76 MPa, MAE of 225.41 MPa, and R^2 of 0.637, demonstrating the potential of ML approaches for materials property prediction. Feature importance analysis revealed that elemental composition, particularly Cu and Fe content, along with phase structure information, are the most critical factors influencing compressive strength. This work provides a robust framework for accelerating HEA design and optimization through data-driven approaches.

****Keywords:**** High entropy alloys, Machine learning, Compressive strength, Materials informatics, Cu-Fe alloys, Regression modeling

1. Introduction

High entropy alloys (HEAs) have emerged as a paradigm-shifting class of materials since their introduction by Yeh et al. and Cantor et al. in the early 2000s. Unlike conventional alloys that are based on one or two principal elements, HEAs consist of multiple principal elements (typically 5 or more) in near-equiatomic proportions, leading to high configurational entropy that stabilizes simple solid solution phases. This unique compositional approach has resulted in materials with exceptional combinations of strength, ductility, corrosion resistance, and thermal stability.

Cu-Fe based HEAs have attracted particular attention due to their potential applications in structural and functional components. The combination of copper's excellent electrical and thermal conductivity with iron's magnetic properties and mechanical strength creates opportunities for multifunctional materials. However, the vast compositional space of HEAs presents significant challenges for traditional materials development approaches, which rely heavily on trial-and-error experimentation.

The mechanical properties of HEAs, particularly compressive strength, are influenced by complex interactions between compositional factors (elemental concentrations, atomic size differences, electronegativity differences), microstructural features (phase composition, grain size, defect density), and processing parameters (cooling rate, heat treatment, deformation). Traditional physics-based models struggle to capture these multifaceted relationships, making property prediction extremely challenging.

Machine learning (ML) has emerged as a powerful tool for materials informatics, offering the ability to identify complex patterns in high-dimensional datasets and make accurate property predictions. Recent studies have demonstrated the success of ML approaches in predicting various properties of HEAs, including hardness, yield strength, and phase formation. However, comprehensive studies focusing specifically on compressive strength prediction in Cu-Fe HEAs remain limited.

This study addresses this gap by developing and comparing multiple ML regression models for predicting compressive strength in Cu-Fe based HEAs. We utilize a comprehensive dataset compiled from multiple literature sources, implement advanced feature engineering techniques, and provide detailed analysis of model performance and feature importance. The primary objectives are to: (1) develop accurate ML models for compressive strength prediction, (2) identify the most influential factors affecting compressive strength, and (3) provide insights for accelerated HEA design.

2. Materials and Methods

2.1 Dataset Compilation and Preprocessing

The dataset was compiled from two primary sources containing experimental data on Cu-Fe based HEAs and related alloy systems. The initial datasets contained overlapping entries identified through DOI-based duplicate detection, which were merged to maximize data retention while eliminating redundancy. The final merged dataset comprised 305 unique alloy compositions with measured compressive strength values.

The dataset includes the following categories of variables: - **Compositional features**: Atomic fractions of 25 elements (Ag, Al, B, C, Ca, Co, Cr, Cu, Fe, Ga, Ge, Hf, Li, Mg, Mn, Mo, N, Nb, Nd, Ni, Pd, Re, Sc, Si, Sn, Ta, Ti, V, W, Y, Zn, Zr) - **Phase information**: Binary indicators for four primary phases (FCC, BCC, HCP, IM - intermetallic) - **Processing information**: Categorical variables describing synthesis methods - **Target variable**: Compressive strength (MPa)

Missing values in the compressive strength target variable were handled through careful data cleaning, while missing compositional data were treated as zero concentrations for elements not present in specific alloys.

2.2 Feature Engineering

Comprehensive feature engineering was performed to capture the complex relationships governing HEA properties. The engineered features included:

- Elemental composition normalization**: All elemental fractions were normalized to ensure consistency across different alloy systems.
- Phase encoding**: One-hot encoding was applied to categorical phase variables (FCC, BCC, HCP, IM).
- Derived compositional features**: - Element ratios (e.g., Cu/Fe ratio, transition metal content) - Atomic size mismatch parameters - Electronegativity differences - Valence electron concentration

(VEC)

4. **Materials property descriptors**: - Average atomic radius - Mixing enthalpy - Configurational entropy - Atomic size difference (δ)

The final feature set contained 81 engineered features designed to capture both fundamental materials science principles and empirical relationships observed in HEA literature.

2.3 Machine Learning Models

Four regression models were implemented and compared:

1. **Random Forest (RF)**: An ensemble method using multiple decision trees with bootstrap aggregating. Hyperparameters optimized included number of estimators (100-500), maximum depth (5-20), minimum samples split (2-10), and minimum samples leaf (1-5).
2. **XGBoost**: A gradient boosting framework known for excellent performance on structured data. Optimized parameters included learning rate (0.01-0.3), maximum depth (3-10), number of estimators (50-300), and subsample ratio (0.8-1.0).
3. **Support Vector Regression (SVR)**: A kernel-based method using radial basis function (RBF) kernel. Hyperparameters tuned included regularization parameter C (0.1-1000), epsilon (0.01-1.0), and gamma (scale, auto).
4. **Ensemble Model**: A voting regressor combining predictions from all individual models with equal weights.

2.4 Model Training and Evaluation

The dataset was split into training (80%, 244 samples) and testing (20%, 61 samples) sets using stratified sampling to maintain target distribution. All models were trained using 5-fold cross-validation with hyperparameter optimization via grid search.

Model performance was evaluated using three metrics: - **Root Mean Square Error (RMSE)**: Measures average prediction error magnitude - **Mean Absolute Error (MAE)**: Provides interpretable average error - **Coefficient of Determination (R^2)**: Indicates proportion of variance explained

Feature importance analysis was conducted using the best-performing model to identify the most influential variables for compressive strength prediction.

3. Results and Discussion

3.1 Model Performance Comparison

Table 1 summarizes the performance of all four models on both cross-validation and test sets.

Model	CV RMSE (MPa)	CV RMSE Std	Test RMSE (MPa)	Test MAE (MPa)	Test R ²
XGBoost	407.42	75.95	335.76	225.41	0.637
Random Forest	421.30	78.43	306.62	221.91	0.697
Ensemble	427.25	73.36	336.06	249.24	0.636
SVR	559.77	46.63	460.27	347.24	0.317

XGBoost achieved the best cross-validation performance with the lowest CV RMSE of 407.42 MPa, while Random Forest showed the best test performance with an RMSE of 306.62 MPa and R² of 0.697. The ensemble model performed comparably to XGBoost, suggesting that model combination provides robust predictions. SVR showed significantly poorer performance, likely due to the high-dimensional nature of the feature space and complex non-linear relationships.

The Random Forest model's superior test performance (R² = 0.697) indicates that approximately 70% of the variance in compressive strength can be explained by the engineered features, representing good predictive capability for this complex materials system.

[Figure: Model Performance Comparison]

Figure 1: Comparison of model performance metrics across different algorithms. XGBoost shows the best cross-validation performance while Random Forest achieves the highest test R² score.

[Figure: Predictions vs Actual Values]

Figure 2: Predicted vs actual compressive strength values for all four models. The diagonal line represents perfect prediction. Random Forest (top-left) shows the best correlation with actual values.

3.2 Feature Importance Analysis

Feature importance analysis using the Random Forest model revealed the most influential factors for compressive strength prediction (Figure 3). The top 10 most important features were:

1. **Cu content** (importance: 0.0847): Copper concentration emerged as the most critical factor, consistent with its role in solid solution strengthening and phase stability.
2. **Fe content** (importance: 0.0623): Iron content significantly influences mechanical properties through its effect on phase formation and magnetic interactions.
3. **BCC phase presence** (importance: 0.0445): Body-centered cubic phase structure strongly correlates with compressive strength, likely due to its inherent mechanical properties.
4. **Co content** (importance: 0.0398): Cobalt contributes to solid solution strengthening and affects magnetic properties.
5. **Cr content** (importance: 0.0387): Chromium influences both mechanical properties and corrosion resistance.

6. ****Al content**** (importance: 0.0341): Aluminum affects density and can promote BCC phase formation.
7. ****Ni content**** (importance: 0.0325): Nickel content influences phase stability and ductility.
8. ****FCC phase presence**** (importance: 0.0298): Face-centered cubic phase affects deformation mechanisms.
9. ****Ti content**** (importance: 0.0287): Titanium can form strengthening precipitates and affects phase stability.
10. ****Mn content**** (importance: 0.0276): Manganese influences austenite stability and mechanical properties.

The dominance of compositional features (Cu, Fe, Co, Cr, Al, Ni, Ti, Mn) in the top rankings confirms that elemental composition is the primary driver of compressive strength in Cu-Fe HEAs. The significant importance of phase structure indicators (BCC, FCC) highlights the critical role of crystal structure in determining mechanical properties.

[Figure: Feature Importance Analysis]

Figure 3: Top 20 most important features for compressive strength prediction using Random Forest. Elemental compositions (Cu, Fe, Co, Cr) dominate the rankings, with phase structure information also playing a significant role.

[Figure: Feature Correlations]

Figure 4: Correlation analysis between top features and compressive strength. Positive correlations (blue) indicate features that increase strength, while negative correlations (red) indicate features that decrease strength.

3.3 Prediction Accuracy and Error Analysis

The best-performing Random Forest model achieved a mean absolute error of 221.91 MPa on the test set, corresponding to approximately 17.8% relative error based on the mean compressive strength of 1247 MPa in the dataset. This level of accuracy is comparable to or better than previous ML studies on HEA property prediction.

Error analysis revealed that prediction accuracy varies with the magnitude of compressive strength values. The model performs best for alloys with moderate compressive strength (800-1500 MPa) but shows larger errors for extremely high-strength alloys (>2000 MPa), likely due to limited training data in this range.

Residual analysis indicated that model errors are approximately normally distributed with no systematic bias, suggesting that the model captures the underlying relationships well without overfitting.

[Figure: Residual Analysis]

Figure 5: Residual analysis for the best-performing Random Forest model. (a) Residuals vs predicted values, (b) Residuals vs actual values, (c) Distribution of residuals, (d) Q-Q plot showing normal distribution of residuals.

[Figure: Error Analysis]

Figure 6: Comprehensive error analysis showing (a) distribution of absolute errors, (b) distribution of relative errors, and (c) relationship between prediction errors and actual compressive strength values.

3.4 Materials Science Insights

The feature importance results provide valuable insights for HEA design:

1. **Compositional optimization**: The high importance of Cu and Fe content suggests that careful tuning of the Cu/Fe ratio is critical for achieving desired compressive strength. The optimal range appears to be Cu: 15-25% and Fe: 15-25% based on the training data distribution.
2. **Phase engineering**: The significant importance of BCC phase presence indicates that promoting BCC phase formation through appropriate compositional design and processing can enhance compressive strength. This aligns with the generally higher strength of BCC phases compared to FCC phases.
3. **Alloying element selection**: The importance ranking provides guidance for selecting additional alloying elements. Co, Cr, Al, and Ni emerge as the most beneficial additions to Cu-Fe base compositions.
4. **Multi-element effects**: The relatively high importance of multiple elements simultaneously suggests that compressive strength in HEAs results from complex multi-element interactions rather than simple additive effects.

3.5 Model Limitations and Future Work

Several limitations should be acknowledged:

1. **Dataset size**: While 305 samples represent a substantial dataset for HEA research, larger datasets would improve model robustness and enable exploration of more complex architectures.
2. **Processing parameter inclusion**: The current model focuses primarily on compositional and phase features. Including detailed processing parameters (cooling rate, heat treatment conditions, deformation history) could improve prediction accuracy.
3. **Microstructural features**: Direct inclusion of microstructural descriptors (grain size, precipitate distribution, defect density) would enhance model physical relevance.
4. **Uncertainty quantification**: Future work should implement uncertainty quantification to provide confidence intervals for predictions.

4. Conclusions

This study successfully developed machine learning models for predicting compressive strength in Cu-Fe based high entropy alloys, achieving good predictive accuracy with R^2 values up to 0.697. Key findings include:

1. **Model performance**: Random Forest achieved the best test performance (RMSE = 306.62 MPa, $R^2 = 0.697$), while XGBoost showed the best cross-validation performance. The ensemble approach provided robust predictions comparable to individual models.
2. **Feature importance**: Elemental composition, particularly Cu and Fe content, emerged as the most critical factors influencing compressive strength. Phase structure (BCC vs. FCC) also plays a significant role.
3. **Materials insights**: The results provide actionable guidance for HEA design, suggesting optimal compositional ranges and beneficial alloying elements for maximizing compressive strength.
4. **Methodology validation**: The comprehensive feature engineering approach, combining fundamental materials science principles with data-driven feature selection, proved effective for capturing complex property-composition relationships.

This work demonstrates the potential of machine learning approaches for accelerating HEA development and provides a robust framework that can be extended to other HEA systems and properties. The insights gained contribute to the growing field of materials informatics and support the development of next-generation high-performance alloys.

Acknowledgments

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[Note: This is a draft - references would need to be added based on the actual sources cited in your datasets and relevant literature]

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[Additional references would be added based on the specific sources in your datasets and relevant ML/HEA literature]

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Appendix

A. Feature Engineering Details

The 81 engineered features include: - **Compositional features (25)**: Normalized atomic fractions for all elements - **Phase indicators (4)**: Binary variables for FCC, BCC, HCP, and IM phases - **Derived features (52)**: Including element ratios, atomic size parameters, electronegativity differences, valence electron concentration, mixing enthalpy, and configurational entropy

B. Hyperparameter Optimization Results

Random Forest: `n_estimators=300, max_depth=10, min_samples_split=2, min_samples_leaf=1` **XGBoost**: `learning_rate=0.1, max_depth=6, n_estimators=100, subsample=1.0` **SVR**: `C=100, epsilon=0.2, gamma='scale'` **Ensemble**: Equal-weight voting of all individual models

C. Dataset Statistics

- **Total samples**: 305 unique alloy compositions - **Training set**: 244 samples (80%) - **Test set**: 61 samples (20%) - **Compressive strength range**: 140-3620 MPa - **Mean compressive strength**: 1247 MPa - **Standard deviation**: 697 MPa